Technology Strategy Report

AI-Powered Forecasting & Automation for SSGC

Transforming Pakistan's largest gas utility through intelligent automation and predictive analytics





Company Overview

Sui Southern Gas Company (SSGC) is Pakistan's largest state-owned utility, responsible for natural gas transmission and distribution across Sindh and Balochistan. Despite advanced SCADA monitoring capabilities, critical operations still rely on manual data collection, introducing inefficiencies and delays.

Core Challenge: Converting operational data into actionable insights through AI-powered forecasting and automation.

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Provinces Served

Sindh & Balochistan

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Key Sectors

Residential, Commercial, Industrial



Current State Analysis

Strengths

- SCADA-based real-time monitoring
- Sole gas provider with stable revenue
- Experienced technical workforce
- Government backing and support

Weaknesses

- Manual data collection processes
- No predictive analytics capability
- Dependence on senior staff
- Limited AI/analytics expertise

Strategic Opportunities & Threats

Opportunities

IoT Integration

Automate data collection across network

Predictive Tools

Optimize supply chain performance

Digital Dashboards

Real-time monitoring and alerts

Workforce Upskilling

Training in analytics and automation

Threats

Cybersecurity Risks

Exposure to hacking and data breaches

Employee Resistance

Unfamiliarity with automation

Vendor Dependency

Reliance on third-party maintenance

Transition Challenges

Data loss or compatibility issues

Proposed Technology Solution

Hybrid Automation & AI-Based Forecasting System



Gas Flow Forecasting

Predict inflow/outflow patterns using historical SCADA data and machine learning models



Inventory Planning

Optimize gas supply scheduling based on demand predictions and seasonal trends



Predictive Maintenance

Minimize equipment failures through AI-driven maintenance scheduling algorithms



Automated Dashboards

Real-time SCADA integration with automated reporting and Excel generation



Strategic Impact

Operational Efficiency

Automation reduces manual workload and improves real-time visibility across the network

Cost Reduction

Predictive maintenance minimizes breakdowns and optimizes repair schedules

Data-Driven Decisions

Forecasting enables proactive planning and faster response to anomalies

Service Reliability

Accurate demand prediction ensures consistent gas supply to all sectors

Implementation Roadmap

Short-Term (0–6 Months)

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- Digitize manual data-entry processes
- Conduct AI and analytics training workshops
- Build initial dashboards with real-time metrics
- Begin pilot forecasting models

Mid-Term (6–18 Months)

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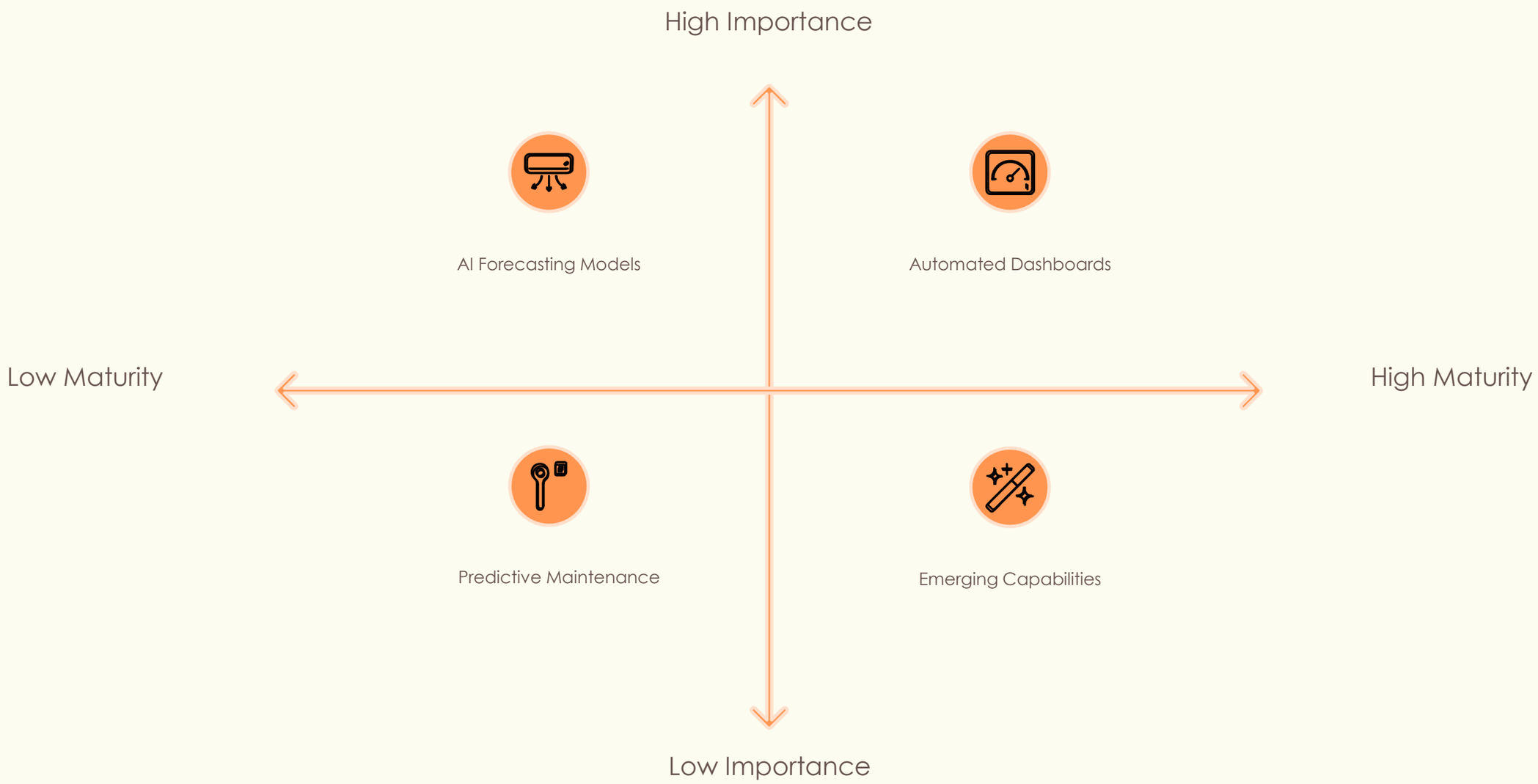
- Fully integrate AI forecasting into SCADA
- Deploy predictive maintenance engine
- Implement smart alerts for anomalies
- Launch automated reporting system

Long-Term (18–36 Months)

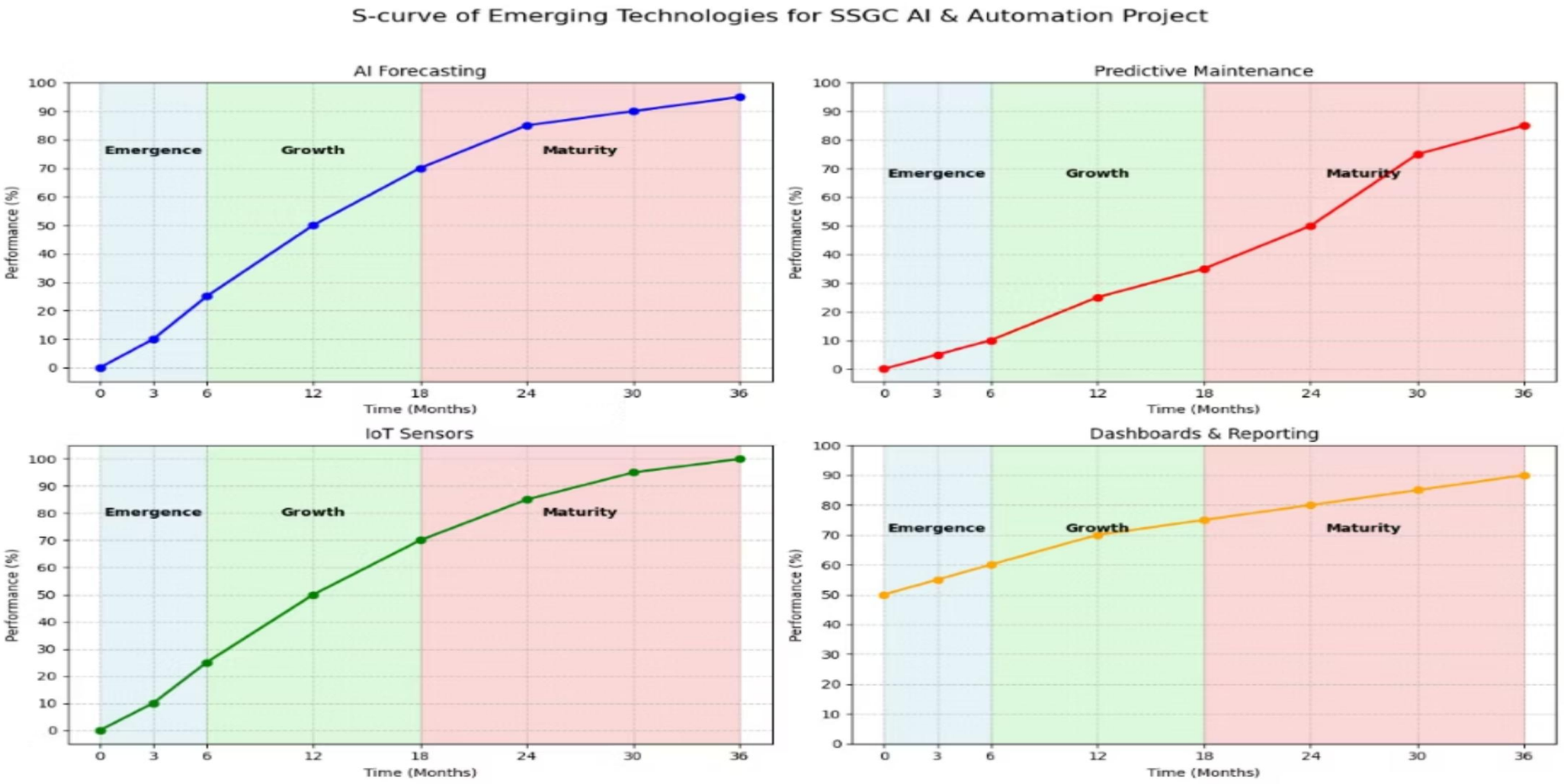
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- Expand forecasting to seasonal trends
- Integrate IoT sensors for remote monitoring
- Automate field operations via mobile capture
- Achieve fully self-adjusting hybrid system

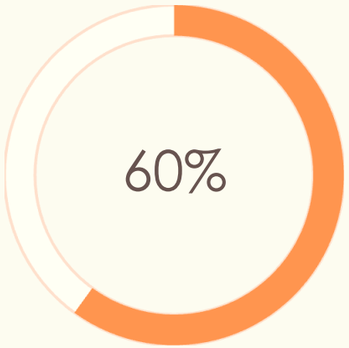
Technology Portfolio & Maturity



Technology S-Curve Analysis

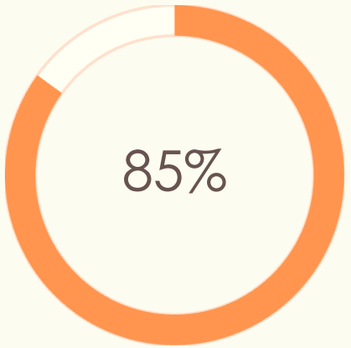


Expected Outcomes & KPIs



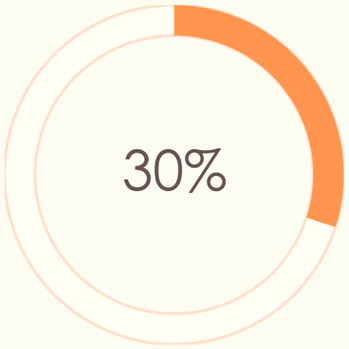
Reduction in Manual Work

Decreased data entry workload



Forecasting Accuracy

Improved prediction reliability



Fewer Failures

Reduction in unplanned maintenance



System Uptime

Increased network reliability

Financial Impact

10–15% reduction in operational costs and 20% decrease in resource wastage

Service Quality

50–70% faster reporting and 40–50% quicker fault response times

Digital Adoption

80% staff trained on AI tools with 70%+ organizational adoption rate



Risk Management & Ethics

Cybersecurity

Implement robust security protocols to protect AI-driven systems from threats and breaches

Workforce Transition

Focus on upskilling rather than replacement; address resistance through training and transparency

Data Integrity

Ensure accuracy in predictive algorithms and maintain strict confidentiality of operational data

IP Protection

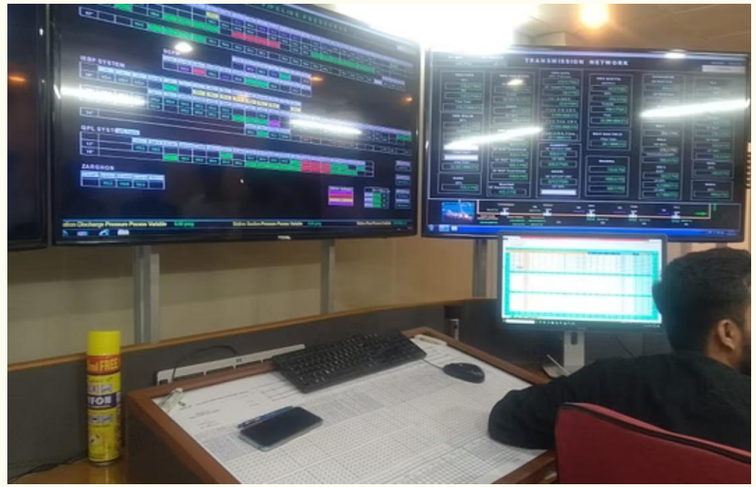
Secure proprietary AI algorithms, dashboards, and maintain data ownership agreements with vendors

SSGC Site Visit & Stakeholder Meeting

As part of our initial engagement phase, a comprehensive site visit was conducted at SSGC facilities. This visit facilitated direct collaboration with key personnel, allowing our team to gain deeper insights into existing operational workflows and infrastructure.







These crucial stakeholder engagement and requirements gathering sessions were instrumental in validating our preliminary assessments and ensuring precise alignment on the strategic objectives for the upcoming AI-powered forecasting and automation initiative.